

Relationships among soil microsite, ectomycorrhizae, and natural conifer regeneration of old-growth forests in western Montana

A. E. HARVEY

Forestry Sciences Laboratory, Intermountain Research Station, United States Department of Agriculture, Forest Service, Moscow, ID, U.S.A. 83843

M. F. JURGENSEN

Department of Forestry, Michigan Technological University, Houghton, MI, U.S.A. 49931

M. J. LARSEN

Center for Forest Mycology Research, Forest Products Laboratory, United States Department of Agriculture, Forest Service, Madison, WI, U.S.A. 53705

AND

R. T. GRAHAM

Forestry Sciences Laboratory, Intermountain Research Station, United States Department of Agriculture, Forest Service, Moscow, ID, U.S.A. 83843

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Successful establishment, root distribution, growth, and ectomycorrhizal development of conifer regeneration in three old-growth forests in western Montana showed site-specific associations with soil microsites containing organic matter. A positive association between decayed wood in the soil and establishment of seedlings occurred on the two drier sites. In general, organic soil components supported most of the root system and ectomycorrhizae on all three sites. Associations between soil organic components and occurrence (establishment) and between organic components and performance (growth) were site specific. No observable evidence of feeder root mortality attributable to soil-inhabiting pathogens was present in any soil component. Roots of competing understory species were notably absent in decayed soil wood.

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Des observations effectuées sur la régénération en conifères dans trois forêts matures de l'ouest du Montana ont montré des associations spécifiques aux sites, en particulier avec les microsites du sol contenant de la matière organique. Sur les deux sites les plus secs, il y a une association positive entre la présence de bois en décomposition dans le sol et l'établissement des plantules. En général et dans les trois sites, les composantes organiques du sol supportent la majeure partie des systèmes racinaires et des mycorhizes. Les associations entre les constituants organiques du sol et la présence (établissement) et entre les constituants organiques et la performance (croissance) sont spécifiques aux sites. Aucune observation n'a révélé la présence de mortalité de racines primaires attribuable à des champignons pathogènes du sol. Les racines d'espèces compétitives d'étages inférieurs étaient remarquablement absentes dans le bois du sol en décomposition.

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A close relationship occurs between ectomycorrhizal development and the ability of conifers to survive and grow in low-fertility forest soils (Bjorkman 1970), on harsh sites (Marx 1976), and in many nursery soils (Trappe and Strand 1969). The failure of afforestation efforts in absence of ectomycorrhizal inoculum (Meyer 1973) demonstrates the importance of well-developed and active ectomycorrhizal root systems for development of vigorous conifers, young or old. Formation of active ectomycorrhizal associations is dependent not only on the presence of fungal inoculum but can also be dependent on a variety of soil chemical and physical components, particularly the content and type of organic matter (Bjorkman 1970; Harvey *et al.* 1979; Kropp 1982; Parke *et al.* 1983). Organic matter and the nutrients contained therein have a direct effect on tree root development (Coutts and Philipson 1977).

Research has shown a positive relationship between organic components of forest soils in the central and northern Rocky Mountains and the ability to support high populations of active ectomycorrhizal root tips in many old-growth forests (Harvey *et al.* 1978, 1980; Harvey *et al.* 1976, 1979). This was particularly

evident under dry soil conditions (Harvey *et al.* 1979). An organic matter relationship may also be important to conifer regeneration. Day (1963) reported large numbers of conifer seedlings established in organic seedbeds relative to mineral seedbeds on harvested sites of the northern Rockies.

Formation of active ectomycorrhizal associations is important for seedling survival and, under some circumstances, is conditioned by organic soil components (Harvey *et al.* 1979; Vogt *et al.* 1981; Parke *et al.* 1983). However, with planted conifer seedlings, most of the ectomycorrhizal activity can occur in mineral soil (Alvarez *et al.* 1979). Other researchers have suggested that the activities of several root pathogens of conifer seedlings may be variously increased or decreased through manipulation of organic materials or as a result of adding them to the rooting medium in forest tree nurseries (Baldwin 1927; Rowe 1955; Wall 1984). Consequently, the availability of organic materials in soil can be either beneficial or deleterious.

Soil organic matter, or lack of it, should affect density, distribution, and growth of conifer seedlings and their root

systems on specific forest sites. This paper examines the potential role of organics on root and ectomycorrhizal distribution from natural conifer regeneration in undisturbed forest soils representative of the northern Rocky Mountains.

Materials and methods

This study was conducted on three sites within the boundaries of the Coram Experimental Forest, between Flathead Lake and Glacier National Park in northwestern Montana. The major timber type in this area is Douglas-fir/larch.

The three study sites represent a range of temperature and moisture conditions typical of the Douglas-fir/larch timber type in northern Rocky Mountain forests. Each supports a mature forest about 250 years of age, with no history of disturbance, and includes at least 1 ha of uniform conditions. Elevations range from 1018 to 1942 m above sea level. Mean annual precipitation at lower elevations averages 78.7 cm and the mean annual temperature is 5.3°C. Mean annual summer temperatures average 16.1°C (Klages *et al.* 1976).

Site 1 is a warm, dry, south slope dominated by Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco). Site 2 is a cool, moist, east slope dominated by Douglas-fir, western larch (*Larix occidentalis* Nutt.), subalpine fir (*Abies lasiocarpa* (Hook.) Nutt.), and Engelmann spruce (*Picea engelmanni* Parry). Western hemlock (*Tsuga heterophylla* (Raf.) Sarg.) dominates site 3, a warm, moist, north slope. These sites represent *Pseudotsuga menziesii*/*Physocarpus malvaceus* (PSME/PHMA), *Abies lasiocarpa*/*Clintonia uniflora* (ABLA/CLUN), and *Tsuga heterophylla*/*Clintonia uniflora* (TSHE/CLUN) habitat types, respectively (Pfister *et al.* 1977). These sites have been characterized in more detail by Harvey *et al.* (1979). Understory vegetation characteristic of the habitat types is described in detail by Pfister *et al.* (1977) and additional details on soils and climate are available in Klages *et al.* (1976).

Each site was sampled with 50 soil cores using the technique of Jurgensen *et al.* (1977). Five cores were taken randomly from around each of 10 evenly dispersed permanent plot centers spaced on a 30-m grid. These cores were used to determine mean soil-component (litter, humus, decayed wood, and mineral) depth (cm) for each site. Natural, established, whole seedlings were also removed and root distribution patterns were determined for the various soil components. Ten seedlings, regardless of species, were taken randomly from around each of the 10 permanent plot centers. Seedlings were taken in late June and early July of 1980, 100 seedlings of various species from each site. Only well-established seedlings that were considered likely to survive were sampled.

Each seedling was lifted with a long-bladed spade, taking great care to prevent damage to the root systems. The root ball was carefully dissected in the field, and the root systems were removed intact. Since understory vegetation was relatively sparse and root systems were not extensive, this process was time consuming but presented no serious problems. The following data were recorded during or subsequent to the root ball dissection process: species, age, height and length of longest root, top and root weight (oven dry), seedbed characteristics at both random and seedling locations (depth of moss, humus, soil wood, and mineral soil to 30 cm, approximately the maximum rooting depth of the seedlings), distribution of the roots in each soil component (percentage estimated of total root length), the particular soil component that supported 50% or more of all ectomycorrhizal short roots, and visual evidence of feeder root pathogens (root lesions and necrosis).

Each seedling was placed in a plastic bag and returned to the laboratory where ectomycorrhizal root tips (0–4, 5–24, 25–49, 50–74, 75–94, and 95–100, as a percentage of total root tips) were estimated and total number of active ectomycorrhizal root tips were counted. All roots were examined for visible, external evidence of activity of feeder root pathogens (root lesions or necrosis). Shoots and roots were subsequently oven dried at 60°C for 24 h and weighed.

Analysis of variance was used for testing the effects of site, soil component, ectomycorrhizae, and root distribution on tree growth. If significant differences were found, Duncan's multiple range test was

TABLE 1. Rooting depth and age of natural seedlings from three undisturbed stands in northwestern Montana

Forest site	Mean age (year)	Mean root depth (cm)
Site 1 (PSME/PHMA)*	4a†	11x
Site 2 (ABLA/CLUN)	11b	15y
Site 3 (TSHE/CLUN)	8b	14y

*Habitat type designation (Pfister *et al.* 1977).

†Differing letters indicate a significant difference (Duncan's $\alpha = 0.05$) among sites.

used to evaluate significance of differences between means. A *t*-test was used for comparing vertical soil profiles from random and seedling locations. A χ^2 analysis was used to compare distributions of seedlings in soil components within and between sites. We considered that seedlings had been established in the respective soil component (litter, humus, decayed wood, mineral) if more than half of the root system occurred within that component.

Results

Although the overstory stands on all three sites were of similar age (ca. 250 years), the natural regeneration on the dry Douglas-fir site (site 1) was significantly younger (mean 4 years) than on sites 2 or 3. The mean rooting depth of the younger seedlings from site 1 was also significantly shorter than that from seedlings on the other two sites. All root systems were well developed though not extensive enough to be difficult to remove from the soil with a ca. 50-cm spade (Table 1).

The organic soil component depths of soil microsites providing satisfactory conditions for establishment of natural conifer seedlings varied significantly when compared with average soil conditions. For example, there were significantly ($\alpha = 0.01$) lower quantities of litter and humus but more decayed wood (not significant) in seedling microsites on the dry Douglas-fir site (site 1), significantly ($\alpha = 0.01$) more litter and decayed wood from seedling microsites on the subalpine fir stand (site 2), and significantly ($\alpha = 0.01$) more litter and humus for seedling microsites from the hemlock stand (site 3) than were represented from average soil component conditions on these sites (Table 2).

Examination of the distribution of seedling root systems revealed a tendency for concentration of the roots and ectomycorrhizae in the organic soil components of all three sites (Tables 3 and 4). There were indications that the distribution patterns might change with different species. However, differences based on species were not significant. When all species were considered together, there were highly significant differences in distribution patterns of seedling root systems among soil components (humus, decayed wood, and mineral), both within and between sites. These distribution patterns indicated a positive relationship between organic soil components and seedling success, particularly decayed wood in the dry Douglas-fir stand (Table 4).

To determine possible relationships between soil components, ectomycorrhizae, and growth, we analyzed the distribution of ectomycorrhizal tips on an equivalent root weight (short root/g root) basis to account for differences in seedling size. Similarly, seedling growth was analyzed as a function of top weight per year (top weight/age) to help account for variation in age among seedlings sampled. On this basis the only significant site–ectomycorrhizal relationships were higher densities on the hemlock site (site 3), especially in decayed wood (Table 5).

TABLE 2. Random organic soil component depths ($n = 50$) compared with organic soil component depths supporting natural conifer seedlings ($n = 100$) from three undisturbed stands in northwestern Montana

Site and soil component	Random profile ($\bar{x}$, cm)	Seedling microsite profile ($\bar{x}$, cm)
Site 1 (PSME/PHMA)*		
Litter	0.1a†	0.6b
Humus	1.3l	0.6m
Decayed wood	0.8x	1.2x
Total	2.2	2.4
Site 2 (ABLA/CLUN)		
Litter	0.3a	1.0b
Humus	1.6l	1.9l
Decayed wood	1.7x	4.5y
Total	3.6	7.4
Site 3 (TSHE/CLUN)		
Litter	0.4a	1.3b
Humus	1.4l	2.9m
Decayed wood	1.9x	1.6x
Total	3.7	5.8

*Habitat type designation (Pfister *et al.* 1977).†Differing letters indicate a significant difference ($\alpha = 0.05$, *t*-test) between random and seedling microsite component depths (*ab*, *lm*, *xy*).

TABLE 3. Percent of natural seedlings (ca. 100)* from each of three undisturbed stands in northwestern Montana that have more than 50% of the ectomycorrhizal root tips in respective soil component

Forest site and seedling species	Soil component			
	Moss	Humus	Decayed wood	Mineral
Site 1 (PSME/PHMA)†				
Douglas-fir	0‡	23	72	4
Western white pine (<i>Pinus monticola</i> Dougl.)	0	0	1	0
Site 2 (ABLA/CLUN)				
Douglas-fir	0	29	21	6
Western white pine	0	0	1	0
Subalpine fir	0	15	20	5
Englemann spruce	0	0	2	0
Western hemlock	0	0	1	0
Site 3 (TSHE/CLUN)				
Douglas-fir	0	9	1	1
Western hemlock	0	37	45	4
Western red cedar	0	0	0	0
Western larch	0	0	1	0

*Some seedlings (32) did not have 50% of their ectomycorrhizal short roots in any one soil component.

†Habitat type designation (Pfister *et al.* 1977).

‡Because of variation in cell sizes and numbers, species differences are not significant; see other tables for seedling differences not related to species.

Considering all soil components, the largest numbers of ectomycorrhizae per gram of root were significantly related to the smallest trees (root, top, or total dry weight), with or without inclusion of nonectomycorrhizal western red cedar as part of the sample population (*Thuja plicata* D. Don.) (ANOVA $\alpha = 0.01$). Considering each soil component separately (smaller sample), this trend was not significant. Differences in growth (top weight/year) were limited to seedlings with root systems primarily in organic matter. Growth rates were generally lower in the dry Douglas-fir stand than in the others. However, the

TABLE 4. Numbers (per plot) of established, natural seedlings from each of three undisturbed stands in northwestern Montana that have more than 50% of the root systems in respective soil component*

Forest site	Soil component		
	Humus	Decayed wood	Mineral
Site 1 (PSME/PHMA)†	20‡	69	8
Site 2 (ABLA/CLUN)	39	42	15
Site 3 (TSHE/CLUN)	50	39	10

*Some seedlings (8) did not have 50% of their root system in any one soil component.

†Habitat type designation (Pfister *et al.* 1977).‡Comparisons between component (rows) and location (columns) are significant, χ^2 , $p = 0.0001$.

TABLE 5. Ectomycorrhizal short roots per gram of root from natural seedlings in three undisturbed stands in northwestern Montana that have more than 50% of the root systems in respective soil component

Forest site	Soil component		
	Humus	Decayed wood	Mineral
Site 1 (PSME/PHMA)*	24ax†	43ax	51ax
Site 2 (ABLA/CLUN)	43ax	55ax	15ax
Site 3 (TSHE/CLUN)	105ay	300by	49ax

*Habitat type designation (Pfister *et al.* 1977).†Different letters indicate significant differences between rows (*a*, *b*) and between columns (*x*, *y*), Duncan's $\alpha = 0.05$.TABLE 6. Mean growth ($\bar{x}$ top dry weight in g/year) of natural seedlings from three undisturbed stands in northwestern Montana that have more than 50% of the root systems in the respective soil component

Forest site	Soil component		
	Humus	Decayed wood	Mineral
Site 1 (PSME/PHMA)*	0.06ax†	0.09ax	0.12ax
Site 2 (ABLA/CLUN)	0.11ay	0.16ay	0.14ax
Site 3 (TSHE/CLUN)	0.12ay	0.08bx	0.15ax

*Habitat type designation (Pfister *et al.* 1977).†Different letters indicate significant differences within rows (*a*, *b*) and within columns (*x*, *y*), Duncan's $\alpha = 0.05$.

slowest growth occurred on seedlings with their root systems primarily in decayed wood of the most productive of the three sites (site 3) (see Table 6).

To segregate soil microenvironmental factors that might contribute differentially to seedling growth and survival, we separated the seedlings from each site into the best performing and least performing half of the populations. This separation into two performance classes showed significant changes in seedling root distribution among soil components within site (Table 7). For the dry Douglas-fir site (site 1), distribution of seedling roots among the soil components and between performance classes was almost identical. On the moderate subalpine fir site (site 2), there were more roots of the better performing seedlings in decayed wood than other soil components. On the mesic hemlock site (site 3), most roots of better performing seedlings were found in the soil humus layers. There was a direct, positive association between the percentage of roots in decayed wood on the dry site (site 1) and growth of the poorest performing seedlings (ANOVA $\alpha = 0.01$).

No visible evidence of pathogen activity was found on feeder

TABLE 7. Numbers of high performing (top 50% based on top weight in g/year) and low performing (bottom 50%) natural seedlings from three undisturbed stands in northwestern Montana that have more than 50% of the root systems in the respective soil component

Forest site	Soil component		
	Humus	Decayed wood	Mineral
High performing			
Site 1 (PSME/PHMA)*	9†	31	6
Site 2 (ABLA/CLUN)	13	30	8
Site 3 (TSHE/CLUN)	28	11	5
Low performing			
Site 1	10	31‡	2
Site 2	25	12	6
Site 3	18	28	4

*Habitat type designation (Pfister et al. 1977).

†Comparisons between components (rows) within sites are significant, χ^2 , $p = 0.0001$.

‡There was a significant positive relationship between this soil component and growth (ANOVA $\alpha = 0.01$).

roots of the seedlings studied. Also, we noted that few roots of any competing understory plants occurred in the decayed wood component of these soils, and on the Douglas-fir site seedlings tended to occur in association with buried logs in the soil.

Discussion

Although a significant difference in age and root length (younger, shorter) occurred with seedlings on the Douglas-fir site, root systems were well enough developed (average length 11 cm) to fully explore the available soil components on the site. Mean depths of the organic components for either random or seedling sites averaged less than 2.5 cm in depth (Table 1). Therefore, we do not feel that age or size differentials affected seedling root distribution patterns of site 1 any differently than the other two sites.

In general, it appears that differences in vertical soil profiles between the random and seedling microsites are most likely related to the specific moisture conditions of each site. For example, in the Douglas-fir stand there was little litter and humus, but substantial amounts of decayed wood associated with seedlings. Because this is an extremely droughty site, the moisture content of the thin layers of litter and humus are likely low most of the growing season. The decayed wood, on the other hand, occurs in sufficient quantity to provide a potential reservoir of moisture throughout the growing season (Harvey *et al.* 1978). A similar relationship probably holds with decayed wood in the subalpine fir site, but moisture is sufficient for litter and humus to be a more favorable rooting medium than on the dry site. In contrast, the litter and humus layers appear to be excellent establishment media on the moist hemlock site, and there seems to be little need for the water provided by decayed wood.

The tendency for seedling roots and associated ectomycorrhizae to concentrate in soil organic components (Tables 3, 4, and 5) is a clear indication that the organic microsite effect is not limited to seedbeds as indicated by Day (1963) for fir, spruce, and pine on cutover and burned sites in the northern Rocky Mountains of Canada. In fact, availability of organic soil components may be a key to early survival, even if they represent a relatively poor seedbed for efficient germination. It is evident from our study that the relationship between seedling success and soil components is highly complex in these three stands. This relationship could be conditioned by species,

although no significance was detected in statistical tests (Table 3). The relationship is clearly conditioned by site (Tables 4 and 5). Decayed wood may be particularly critical, at least to early seedling success, on the dry site (site 1).

The relationships between soil components, ectomycorrhizae, and seedling growth were also complex and somewhat surprising. Ectomycorrhizal development was relatively non-responsive to soil components in a statistical sense, except for the high numbers of short roots in decayed wood of the hemlock stand. High variability in ectomycorrhizal root tips, perhaps caused by presampling moisture history, limited our ability to discern other than strong interactions. In fact, the large numbers of short roots in the decayed wood of the hemlock stand (site 3) and relatively poor growth rate of seedlings in this stand yielded a significantly negative relationship between large numbers of ectomycorrhizae and tree size. Because the forest floor of the hemlock stand is well shaded, the carbon reserves (photosynthetic ratio) of seedlings may be more important to growth than are the nutrients or water provided by ectomycorrhizae. The difference in growth rates between the Douglas-fir stand seedlings (site 1) and seedlings in the other stands is probably the result of smaller size and drier site conditions.

Separating seedling data by performance (growth) class to segregate survival versus growth effects was based on the assumption similar patterns between classes would indicate similar requirements and differing patterns would indicate different requirements for establishment and good performance. Because root distribution patterns on the dry Douglas-fir site were nearly identical for both performance classes, the soil requirements for both establishment and growth are likely similar. The strong positive association between the presence of decayed wood and growth of the low performance class is indicative that this component is particularly important to seedling growth on this harsh site. Decayed wood in soils of these sites retains moisture even during the summer draught periods (Harvey *et al.* 1978; Harvey *et al.* 1979).

In contrast, patterns of root distribution among soil organic components changed between seedling performance classes on the other two sites, suggesting a different interpretation. In the subalpine fir site, humus was apparently important for establishment and decayed wood for growth. With the hemlock site, decayed wood was indicated as important to survival and humus important to growth.

Although the aboveground components of these ecosystems (sites 1, 2, and 3) may have affected results, conditions were quite uniform in these stands, i.e., no large openings created by either natural or man-caused events. Thus, the aboveground physical environment likely had little effect. On the other hand, belowground competition with roots from the overstory was likely significant. Previous studies have shown that distribution of ectomycorrhizal activities from the overstory on these sites is similar to that of the seedlings (Harvey *et al.* 1976, 1979; Harvey *et al.* 1978). Thus, seedlings were in direct competition for the same soil space as the overstory. Even with the competition, it is apparently advantageous for feeder roots and ectomycorrhizae of seedlings under these conditions to concentrate in organic layers.

We found no root lesions, general necrosis, or other visual evidence of feeder root pathogen activity of these seedlings. Apparently such pathogens have little or no effect on root distribution of established seedlings on these sites. However, there may have been effects early in the establishment process, particularly shortly after germination, as reported elsewhere (Baldwin 1927; Rowe 1955).

Conclusion

A strong site-specific relationship has been demonstrated between soil microsite, particularly organic components, natural seedling establishment, growth, and distribution of seedling ectomycorrhizae in natural forest stands. These relationships may be conditioned by seedling species and climatic conditions. Interactions were complex and are likely mediated primarily by soil moisture. Shading from the overstory likely had a major influence on growth – soil component interactions, particularly in the dense hemlock stand. Lack of roots from competing vegetation may also be a contributing factor for response of seedlings with root systems concentrated in the decayed wood soil component.

Application

Stands in this study are not representative of situations normally considered reforestation problems. However, management to obtain advance regeneration depends on seedlings started under established stands. Also, initiating natural regeneration of these species in some partial cut situations (heavy shading) may be dependent on somewhat similar conditions. Because of the highly "reactive" nature of the seedling establishment–growth interaction and soil organic components, and the similarity between distribution of roots and ectomycorrhizae of both seedlings and older stands (Vogt *et al.* 1981; Harvey *et al.* 1980), we recommend protecting these materials in harvesting operations. Where site preparation is necessary to achieve adequate establishment of regeneration, either planted or natural, we recommend that disturbance, in the form of massive break-up of organic layers, be minimized and that a maximum diversity of microsites be retained. This should allow for at least a portion of the microsites to be suitable for a variety of seedbed, rooting media, and ectomycorrhizal requirements to be met.

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